

CENG 515

Numerical Methods for Computer Graphics

Instructors:

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Schedule:

- Monday 13:30

Course Description:

In this course, recent advances in computer graphics and fundamental topics in numerical methods for computer graphics are discussed.

Course Objectives:

1. To examine inspiring topics in computer graphics
2. To explore open subjects in popular research fields in computer graphics
3. To obtain the basic literacy in numerical methods

Reference Books:

Numerical Recipes – The Art of Scientific Computing – 3rd Edition by William H. Press, Saul A. Teukolsky, William T. Vetterling and Brian P. Flannery.

Numerical Algorithms – Methods for Computer Vision, Machine Learning, and Graphics by Justin Solomon.

Numerical Methods for Scientists and Engineers by Richard W. Hamming.

Reading:

Hart, J. Sphere tracing: a geometric method for the antialiased ray tracing of implicit surfaces. *The Visual Computer* **12**, 527–545 (1996).

<https://doi.org/10.1007/s003710050084>

Tentative Schedule:

Week		
1	Introduction	Reading – Sphere Tracing
2	Root finding methods	
3	Introduction to Shaders	HW – Newton Fractals
4	Implicit Surfaces, Signed Distance Fields	

5	Sphere Tracing	
6	Computer Graphics recap	HW – Sphere Tracing
7	Interval Arithmetic	
8	Introduction to Neural Implicits	Reading – Spelunking in the deep
9	Fast marching methods	
10	Intro to Monte Carlo Methods – pde estimation	Reading – Monte Carlo PDE Keenan Crane
11	Presentations	

Suggested Papers for Presentation:

John C. Hart (1996). Sphere tracing: a geometric method for the antialiased ray tracing of implicit surfaces. *The Visual Computer*, 12, 527-545.

Galin, E., Guérin, E., Paris, A. and Peytavie, A. (2020), Segment Tracing Using Local Lipschitz Bounds. *Computer Graphics Forum*, 39: 545-554.
<https://doi.org/10.1111/cgf.13951>

Sawhney, R., & Crane, K. (2020). Monte Carlo geometry processing: a grid-free approach to PDE-based methods on volumetric domains. *ACM Trans. Graph.*, 39(4).

Vincent Sitzmann, Julien N. P. Martel, Alexander W. Bergman, David B. Lindell, & Gordon Wetzstein (2020). Implicit Neural Representations with Periodic Activation Functions. *CoRR*, abs/2006.09661.

Park, J., Florence, P., Straub, J., Newcombe, R., & Lovegrove, S. (2019). DeepSDF: Learning Continuous Signed Distance Functions for Shape Representation. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Sharp, N., & Jacobson, A. (2022). Spelunking the deep: guaranteed queries on general neural implicit surfaces via range analysis. *ACM Trans. Graph.*, 41(4).

IZTECH Department of Computer Engineering Generative Artificial Intelligence Policy

At the Department level, the use of generative artificial intelligence is divided into three main categories.

A. Permitted Use

Unless otherwise specified in the specific policy of the relevant course, the following uses may generally be considered as supportive use:

- requesting conceptual review and explanations of topics;
- summarizing course notes;
- preparing a personal study plan;
- receiving assistance on code syntax;
- interpreting error messages;
- receiving support for writing, grammar, and clarity of expression;
- receiving support for bibliography formatting;
- asking for a topic to be explained through different examples;
- generating sample questions for exam preparation;
- requesting alternative explanations as part of the personal learning process.

However, such uses do not remove the requirement that the final product submitted for assessment reflect the student's own effort, judgment, and academic contribution.

B. Conditionally Permitted Use

The following uses are acceptable only if they are explicitly permitted in the course syllabus appendix and if the required disclosure is provided:

- improving the draft wording of an assignment;
- writing code fragments with the assistance of generative artificial intelligence;
- generating code for experimental analysis;
- preparing data visualization code;
- linguistically editing certain sections of a project report;
- receiving preliminary suggestions for a literature review;
- creating the draft structure of presentation slides;
- receiving ideas on model, algorithm, or experimental design;
- generating test scenarios or unit tests;

- receiving AI-supported explanations during the debugging process.

In such cases, the student must clearly disclose the name of the tool used, the model/version information, the purpose of use, the sections in which it was used, and the extent to which it influenced the final product.

C. Prohibited Use

Unless explicitly permitted by the course instructor, the following uses shall be considered violations of academic integrity:

- having artificial intelligence write the entirety or a substantial part of an assignment;
- receiving assistance from an AI tool during an examination, quiz, or individual assessment;
- having artificial intelligence generate the main algorithm in a coding assignment and presenting it as one's own work;
- having an AI tool write the original analysis, discussion, or conclusion sections of a project report;
- submitting a mathematical solution, proof, or algorithmic analysis as AI-generated output;
- using AI output that includes fabricated sources or unverified references;
- submitting AI-generated text, code, visuals, tables, or analyses without disclosure;
- assigning artificial intelligence an academic task that would be inappropriate to ask another person to perform.

An academic task that would be inappropriate for a student to ask another person to perform is likewise inappropriate to assign artificial intelligence.